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Forums » Commercial and US Government Launch Vehicles » Boeing Starliner (CST-100) Section » Boeing's Starliner (CST-100) - Discussion Thread 4



Poll

Should NASA require Boeing to repeat its Starliner Orbital Test Flight due to the recent software issues?

Yes	460 (95.6%)
No	21 (4.4%)

Total Members Voted: 481

Voting closed: 02/21/2020 10:34 pm

Pages: [**1**] 2 3 ... 80 Next [Go Down](#)

Quick Search

Author

 **Chris Bergin**

NSF Managing Editor
Administrator
Senior Member



Posts: 146419



Liked: 144344
Likes Given: 25918

Topic: Boeing's Starliner (CST-100) - Discussion Thread 4 (1)

[Boeing's Starliner \(CST-100\) - Discussion Thread 4](#)

« on: 04/16/2019 03:22 pm »

New Discussion Thread for Boeing's Starliner/CST-100.

Thread 1:

<https://forum.nasaspaceflight.com/index.php?topic=22125.0>

Thread 2:

<https://forum.nasaspaceflight.com/index.php?topic=32438.0>

Thread 3:

<https://forum.nasaspaceflight.com/index.php?topic=39244.0>

News articles:

<https://www.nasaspaceflight.com/tag/cst-100/>

<https://www.nasaspaceflight.com/tag/starliner/>

L2 Master Thread:

<https://forum.nasaspaceflight.com/index.php?topic=29664.0>

--

Civil and respectful at all times and remain on topic please.

 clongton likes this

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☐ Rocket Science

Senior Member



Posts: 10586

NASA Educator Astronaut
Candidate Applicant
2002

Liked: 4553

Likes Given: 13523

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« Reply #1 on: 04/23/2019 02:54 pm »

Quote from: Chris Bergin on 04/23/2019 02:49 pm

In the water.

Capsule in the water. #Starliner #CommercialCrew #NASA @NASASpaceflight pic.twitter.com/QBMFpSLmf1

— Chris G - NSF (@ChrisG_NSF) April 23, 2019

https://twitter.com/ChrisG_NSF/status/1120699843444465664

Somewhat amusing that Boeing has told the assembled media if this test doesn't go to plan, they aren't at anything!

They really can't get their media message right at all.

I guess they're new on the aerospace scene... 🤔

 david1971 and guyw like this

"The laws of physics are unforgiving"

~Rob: Physics instructor, Aviator

☒ Chris Bergin

NSF Managing Editor
Administrator
Senior Member



Posts: 146419



Liked: 144344

Likes Given: 25918

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« Reply #2 on: 04/23/2019 05:24 pm »

Boeing says it was the 45th or concerns relating the 45th's accounts - worried about negative Boeing worried about unflattering pics of the boilerplate, etc. etc.

A correction here. While it was communicated to media at event that we weren't to take pictures when wrong, @BoeingSpace has reached to clarify that this was a request for 4th not media. All is well. Boeing wasn't trying to hide anything. <https://t.co/q8KY1OtSAA>

— Chris G - NSF (@ChrisG_NSF) April 23, 2019

https://twitter.com/ChrisG_NSF/status/1120739794714279936

« Last Edit: 04/23/2019 05:25 pm by Chris Bergin »

 JEF_300 likes this

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☐ whitelancer64

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« Reply #3 on: 05/16/2019 03:09 pm »

Senior Member



Posts: 4953
Duarte, California, USA



Liked: 4355
Likes Given: 17528

☐ **Bubbinski**

I am an "all space" fan
Senior Member



Posts: 3340
Utah/Arizona



Liked: 736
Likes Given: 402

☐ **Joseph Peterson**

Full Member



Posts: 752
Pittsburgh, PA
Liked: 578
Likes Given: 14356

☐ **OM72**

Full Member



Posts: 146
Houston, TX
Liked: 138
Likes Given: 0

Does anyone know the breakdown in mass between the Starliner capsule and its service m

"One bit of advice: it is important to view knowledge as sort of a semantic tree -- make sure you understand principles, ie the trunk and big branches, before you get into the leaves/details or there is nothing for them Musk

"There are lies, damned lies, and launch schedules." - Larry J

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« **Reply #4 on:** 05/24/2019 12:23 am »

Here's a local angle regarding Starliner landings:

<https://www.ksl.com/article/46558269/astronaut-landings-in-utah-very-likely-and-soon>

I'll even excitedly look forward to "flags and footprints" and suborbital missions. Just fly...somewhere.

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« **Reply #5 on:** 05/24/2019 05:19 pm »

Quote

WASHINGTON — Boeing has completed ground testing of the thrusters for its CST-100 Starliner commercial spacecraft nearly a year after a setback in earlier testing of those thrusters.

In a statement, Boeing said it completed hot-fire testing May 23 of the spacecraft's entire propulsion system, including various thrusters, fuel tanks and related systems within a "flight-like" service module of the spacecraft. The tests took place at NASA's White Sands Test Facility in New Mexico.

A series of tests demonstrated thruster firings for in-space maneuvers, high-altitude aborts and low-altitude landings, the company said the tests were all successful.

<https://spacenews.com/boeing-completes-tests-of-starliner-thrusters/>

First, great to hear the tests were all successful.

Can someone clarify what "flight-like" means?

Jimmy_C and JEF_300 like this

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« **Reply #6 on:** 05/24/2019 06:10 pm »

Quote from: Joseph Peterson on 05/24/2019 05:19 pm

Can someone clarify what "flight-like" means?

It simply means the SM tested was the design intended for flight.

 Joseph Peterson and whitelancer64 like this

 SWGlassPit

I break space hardware
Full Member



Posts: 862
Liked: 930
Likes Given: 146

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« **Reply #7 on:** 05/24/2019 07:24 pm »

As someone who does lots of destructive testing (hypervelocity impact testing for meteoroid debris work), we use both analog and flight-like articles.

Analog test articles are inexpensive approximations to the flight configuration that we use understanding of the test behavior. When we think we're zeroed in on the right size shot, v flight-like hardware to confirm that the approximations made for the analog hardware don't any surprises when applied to the more expensive hardware.

 Coastal Ron, JEF_300, Joseph Peterson and 1 other like this

 otlski

Full Member



Posts: 113
Liked: 209
Likes Given: 207

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« **Reply #8 on:** 05/25/2019 11:18 pm »

Quote from: whitelancer64 on 05/16/2019 03:09 pm

Does anyone know the breakdown in mass between the Starliner capsule and its service module?

SM dry mass = 10000 lbs.

SM wet mass = 15000 lbs

CM dry mass = 14000 lbs

CM wet mass = 18000 lbs

 whitelancer64 and JEF_300 like this

 envy887

Senior Member



Posts: 8569
Liked: 7381
Likes Given: 3032

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« **Reply #9 on:** 05/27/2019 10:06 pm »

Quote from: otlski on 05/25/2019 11:18 pm

Quote from: whitelancer64 on 05/16/2019 03:09 pm

Does anyone know the breakdown in mass between the Starliner capsule and its service module?

SM dry mass = 10000 lbs.

SM wet mass = 15000 lbs

CM dry mass = 14000 lbs

CM wet mass = 18000 lbs

Wow. That's a lot. The N22 can get that to orbit?

LSP says that is 541 territory.

« Last Edit: 05/27/2019 10:08 pm by envy887 »

☐ **Zed_Noir**

Senior Member



Posts: 5490

Canada

Liked: 1815

Likes Given: 1302

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« Reply #10 on: 05/27/2019 11:24 pm »

Quote from: envy887 on 05/27/2019 10:06 pm

Quote from: otlski on 05/25/2019 11:18 pm

Quote from: whitelancer64 on 05/16/2019 03:09 pm

Does anyone know the breakdown in mass between the Starliner capsule and its service module?

SM dry mass = 10000 lbs.

SM wet mass = 15000 lbs

CM dry mass = 14000 lbs

CM wet mass = 18000 lbs

Wow. That's a lot. The N22 can get that to orbit?

LSP says that is 541 territory.

The Atlas V 541 have to deal with gravity loss with the puny single Centaur when orbiting its payload to LEO. According to wikipedia the Atlas V 402 can get 12500 kg (27557 lb) to LEO SRB should give adequate margins to orbit the Starliner stack.

☐ **Steven
Pietrobon**

Member

Senior Member



Posts: 40503

Adelaide, Australia



Liked: 34652

Likes Given: 12796

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« Reply #11 on: 06/21/2019 07:35 am »

From the GAO report. New information is that four of the eight valves failed to close in the uncrewed flight is flying with the faulty valves, since the abort system will be disabled.

"In June 2018, Boeing experienced an anomaly while testing its launch abort engines. During four of the eight total valves in the four launch abort engines failed to close after a shutdown was sent. In response to this event, Boeing initiated an investigation to identify the root cause. According to Boeing officials, Boeing plans to replace components on all of its service modules except uncrewed test flight service module. This is because the abort system will not be active for test flight. Boeing plans to resume testing its launch abort engines in May 2019. A NASA of addressing this anomaly and identifying its root cause resulted in a 12-month schedule delay of abort propulsion system testing."

 zubenelgenubi, JEF_300, Joseph Peterson and 2 others like this

Akin's Laws of Spacecraft Design #1: Engineering is done with numbers. Analysis without numbers is only

☐ **Steven
Pietrobon**

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« Reply #12 on: 06/21/2019 07:40 am »

Member
Senior Member



Posts: 40503
Adelaide, Australia



Liked: 34652
Likes Given: 12796

Think its worth repeating all these program risks from the GAO report.

• **Parachute System Certification.** Boeing is conducting five parachute system qualification tests to demonstrate that its system meets the Commercial Crew Program's requirements, which will be conducted on two spacecraft flight tests. However, in August 2018, Boeing identified a faulty release mechanism on its drogue parachute—which initially slows down the capsule—during its third parachute test. The parachute that successfully deployed all parachutes. Identifying and fixing the faulty mechanism delayed the parachute qualification test. According to a NASA official, Boeing is conducting testing to qualify an alternative design, and Boeing must qualify this alternative design before the crewed test flight.

• **Launch Vehicle Engine Anomaly.** Boeing is addressing a safety risk related to a launch vehicle engine component. Specifically, during a 2018 launch, the launch vehicle engine position during a launch sequence from commands but the launch vehicle provider stated that it achieved all mission objectives. Officials told us that they have insight into the launch vehicle manufacturer's ongoing investigation and have participated in a separate independent review team. Boeing will implement a set of corrective actions for the uncrewed test flight, and will continue testing the engines for the crewed test flight.

• **Spacecraft-Generated Debris.** Boeing is addressing a risk that under normal operating conditions, initiators that trigger separation events, such as the separation of the crew and service module during entry, may generate debris and damage the spacecraft. These components function as explosive bolts. Boeing plans to install hardware to contain debris generated when the initiators fire. Program officials told us that they believe Boeing has identified a solution that will be sufficient for the uncrewed test flights, but the program is continuing to explore a possible redesign for future operations.

• **Spacecraft Forward Heat Shield.** We had previously found that Boeing was addressing a risk that during descent a portion of the spacecraft's forward heat shield may re-contact the spacecraft and damage the parachute system. Since our last report, Boeing tested the performance of the forward heat shield in worst-case scenarios and found there was no damage to the parachute or the spacecraft. After reviewing test data, the program determined that Boeing had completed mitigation activities and, as of February 2019, no additional steps were needed."

 [guyw](#), [tyrred](#), [zubenelgenubi](#) and 9 others like this

Akin's Laws of Spacecraft Design #1: Engineering is done with numbers. Analysis without numbers is only

☐ **Lars-J**

Senior Member



Posts: 6832
California
Liked: 8552
Likes Given: 5511

Re: Boeing's Starliner (CST-100) - Discussion Thread 4
« **Reply #13 on:** 08/22/2019 02:28 am »

From the update thread:

Quote from: ChefwithaCamera on 08/08/2019 09:19 pm

You might be wondering how they are going to achieve such an ambitious time goal when it seemed to take things together for the first one?

Well, at the AEHF-5 Social we got a peek inside the Atlas Spaceflight Operations Center (ASOC) where they are testing boosters before sending them out to the Vertical Integration Facility (VIF). It turns out ULA already has the Atlas V core and Dual Centaur upper stage for the Crewed Flight Test. They are currently sitting inside the Vehicle Assembly Building, hopeful launch before the end of 2019.

Well nobody expected ULA to be the long pole for this anyway, so it doesn't mean much. W

wondering is will the spacecraft itself meet the schedule?

TripleSeven

Full Member



Posts: 1145

Istanbul Turkey and
Santa Fe TEXAS USA

Liked: 588

Likes Given: 2094

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« Reply #14 on: 08/24/2019 10:36 am »

Quote from: Steven Pietrobon on 06/21/2019 07:40 am

Think its worth repeating all these program risks from the GAO report.

"• **Parachute System Certification.** Boeing is conducting five parachute system qualification tests to determine if its system meets the Commercial Crew Program's requirements, which will be validated on two spacecraft. However, in August 2018, Boeing identified a faulty release mechanism for its drogue parachute—which is used to pull down the capsule—during its third parachute qualification test that successfully deployed all parachutes. In fixing the faulty mechanism delayed its fourth parachute qualification test. According to a NASA official, Boeing is conducting testing to qualify an alternative design, and Boeing must qualify this alternative design before flight.

• **Launch Vehicle Engine Anomaly.** Boeing is addressing a safety risk related to a launch vehicle component. Specifically, during a 2018 launch, the launch vehicle engine position during ascent deviated from command. The launch vehicle provider stated that it achieved all mission objectives. Program officials told us that they have been monitoring the launch vehicle manufacturer's ongoing investigation and have participated in a separate independent investigation. Boeing will implement a set of corrective actions for the uncrewed test flight, and will continue testing the crewed test flight.

• **Spacecraft-Generated Debris.** Boeing is addressing a risk that under normal operating procedures the separation of the crew and service module prior to re-entry, may generate debris that could trigger separation events, such as the separation of the crew and service module prior to re-entry, may generate debris and damage the spacecraft. These components function as expected, but Boeing plans to install hardware to prevent debris generated when the initiators fire. Program officials told us that they believe Boeing has identified a mitigation that will be sufficient for the uncrewed and crewed test flights, but the program is continuing to explore a possible mitigation for future operational missions.

• **Spacecraft Forward Heat Shield.** We had previously found that Boeing was addressing a risk that during re-entry, a portion of the spacecraft's forward heat shield may re-contact the spacecraft after it is jettisoned and damage the parachute system. Since our last report, Boeing tested the performance of the forward heat shield in worst-case scenarios and found there was no damage to the parachute system or the spacecraft. After reviewing test data, the program determined that Boeing had completed the mitigation activities and, as of February 2019, no additional steps were needed."

this is how an aviation company charge with flying people safely does things. Boeing did not know it would take this long, but they misjudged the Dreamliner and it is (not only a great flying plane but also) the aviation technology of the future. Starliner is the same thing...

it takes some time but well its flight safety software companies issue updates after blue sky issues whatever they are called now as the pieces of disaster are picked up



« Last Edit: 08/24/2019 10:48 am by TripleSeven »

TripleSeven

Full Member



Posts: 1145

Istanbul Turkey and
Santa Fe TEXAS USA

Liked: 588

Likes Given: 2094

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« Reply #15 on: 08/24/2019 11:20 am »

Quote from: Lars-J on 08/22/2019 02:28 am

From the update thread:

Quote from: ChefwithaCamera on 08/08/2019 09:19 pm

You might be wondering how they are going to achieve such an ambitious time goal when it seemed to

to get things together for the first one?

Well, at the AEHF-5 Social we got a peek inside the Atlas Spaceflight Operations Center (ASOC) where the boosters before sending them out to the Vertical Integration Facility (VIF). It turns out ULA already has the assembled Atlas V core and Dual Centaur upper stage for the Crewed Flight Test. They are currently sitting at ASOC awaiting a hopeful launch before the end of 2019.

Well nobody expected ULA to be the long pole for this anyway, so it doesn't mean much. What people are the spacecraft itself meet the schedule?

You did ask a question...so I answer

when Boeing Test Aviators Experimental, not computer geeks on a keyboard who have no experience in the air or much past the keyboard and in the flight sit safely on the ground

say it is ready ...safe flights

« Last Edit: 08/24/2019 11:27 am by TripleSeven »

☐ **Lars-J**

Senior Member



Posts: 6832
California

Liked: 8552

Likes Given: 5511

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« **Reply #16 on:** 08/24/2019 02:59 pm »

Quote from: TripleSeven on 08/24/2019 11:20 am

Quote from: Lars-J on 08/22/2019 02:28 am

Well nobody expected ULA to be the long pole for this anyway, so it doesn't mean much. What people are the spacecraft itself meet the schedule?

You did ask a question...so I answer

when Boeing Test Aviators Experimental, not computer geeks on a keyboard who have no experience in the air or much past the keyboard and in the flight sit safely on the ground

say it is ready ...safe flights

Huh? This is the most bizarre post you have made on this forum.



octavo, jimmy_C, swampcat and 2 others like this



FutureSpaceTourist

Global Moderator
Senior Member



Posts: 57751

Re: Boeing's Starliner (CST-100) - Discussion Thread 4

« **Reply #17 on:** 08/27/2019 06:03 am »

Talk from April 2018 just posted:

UK



Liked: 94848

Likes Given: 44765

NEAF Talks



Watch on

☐ **Vattedrmr**

Senior Member



Posts: 2204

Hot Springs, AR

Liked: 2934

Likes Given: 4845

Re: Boeing's Starliner (CST-100) - Discussion Thread 4« **Reply #18** on: 09/08/2019 11:15 am »

Quick question I haven't found an answer to: When will the Ground Abort test sequence in? After OFT but before CFT? For some reason I'm just not finding anything.

TIA, and have a good one,
Mike

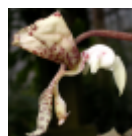
JEF_300 likes this

Aviation/space enthusiast, retired control system SW engineer, doesn't know anything!

☐ **gongora**

Global Moderator

Senior Member



Posts: 10901

US

Liked: 15247

Likes Given: 6767

Re: Boeing's Starliner (CST-100) - Discussion Thread 4« **Reply #19** on: 09/08/2019 01:08 pm »

Between OFT and CFT seems to be the current plan.

JEF_300 and Vattedrmr like this

Tags:

Forums » Commercial and US Government Launch Vehicles » Boeing Starliner (CST-100) Section » Boeing's Starliner (CST-100) - Discussion Thread 4

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